

IRB #: IRB-FY2026-340
Title: Guided Self-directed Learning via LLMs and The Role of The Instructor
Creation Date: 6-5-2026
End Date:
Status: Under Review
Principal Investigator: Iyadunni (Dunni) Adenuga
Review Board: Institutional Review Board #1
Sponsor:

Study History

Submission Type	Initial	Review Type	Exempt	Decision
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Key Study Contacts

Member	Iyadunni (Dunni) Adenuga	Role	Principal Investigator	Contact	iadenuga@kean.edu
Member	Kiana Becca Nunez	Role	Primary Contact	Contact	beccanuk@kean.edu
Member	Iyadunni (Dunni) Adenuga	Role	Primary Contact	Contact	iadenuga@kean.edu
Member	Md Jonayed Hossain Chowdhury	Role	Investigator	Contact	chowdmdj@kean.edu
Member	Kiana Becca Nunez	Role	Investigator	Contact	beccanuk@kean.edu

INSTRUCTIONS

To begin your application, determine the type of review needed based on categories presented below. When prompted, provide the required information including project personnel, study information, and any supporting documents needed for IRB review.

For example, these items must also be submitted (when applicable) with the application for it to be considered complete:

- Assent form (for participants under 18)
- Consent form
- An additional Consent form for participants being photographed or recorded via digital media
- Debriefing form
- Site permission
- Copies of all survey instruments, interview questions, recruitment letters, emails, and/or advertisements
- Permission to use materials created by a different researcher

All Forms must be created using Kean letterhead.

Complete answers to all questions in this application must be provided.

Applications must be certified by all Principal and Co-Principal Investigators.

Incomplete applications will be returned without review.

Direct questions about the IRB application and review process to irb@kean.edu or visit the IRB website at

<https://www.kean.edu/offices/research-and-sponsored-programs/irb-researchcompliance>

REVIEW TYPE DETERMINATION

The following categories will help determine the review type (Exempt, Expedited, or Full board) needed for your submission.

Once a category has been selected, only the sections/questions pertinent to your application type will display on this submission form.

If you do not select a category for Exempt or Expedited review, your application will default to a Full board review.

After submission, all applications are screened by the IRB administrative staff prior to assignment for review by members of the IRB.

EXEMPT CATEGORIES

Please read each category carefully. Click on "More Information" to read additional instructions regarding each category. There are 8 categories in the Exempt Category Group. In order for your research to be applicable under this group, at least one of the categories must apply. **No research involving children or individuals from vulnerable populations is eligible under the Exempt Category Group.**

XM1. Research conducted in established or commonly accepted educational settings involving normal educational practices that is not likely to have adverse impacts on students learning, required educational content, or involving assessment of educators who provide instruction.

[More Information](#)

XM2. Research involving the use of educational tests (cognitive, diagnostic, aptitude, achievement), survey procedures, interview procedures or observation of public behavior.

[More Information](#)

XM3. Research involving benign behavioral interventions in conjunction with the collection of information from an adult subject through verbal or written responses (including data entry) or audiovisual recording if the subject prospectively agrees to the intervention and information collection.

✓ **More Information**

At least one of the following criteria must be met:

1. The information obtained is recorded by the investigator in such a manner that the identity of the human subjects cannot readily be ascertained or through identifiers linked to the subjects.
2. Any disclosure of the human subjects' responses outside the research would not reasonably place the subjects at risk of criminal or civil liability or be damaging to the subjects' financial standing, employability, or educational advancement.
3. The information obtained is recorded by the investigator in such a manner that the identity of the human subjects can readily be ascertained, directly or through identifiers linked to the subjects, and an IRB conducts a limited IRB review to make the determination required by 46.111(a)(7).

IMPORTANT: Deception is allowed if certain criteria are met. This exemption is only for benign behavioral research with adults and is not applicable to children.

For the purpose of this category, benign behavioral interventions are brief in duration, harmless, painless, not physically invasive, not likely to have a significant adverse lasting impact on the subjects, and the investigator has no reason to think the subjects will find the interventions offensive or embarrassing. Provided all such criteria are met, examples of such benign behavioral interventions include having subjects:

- Play an online game
- Solve puzzles under various noise conditions
- Decide how to allocate a nominal amount of received cash between themselves and someone else

The methods of data collection allowed under this category are limited to verbal or written responses from subjects (e.g. surveys or interviews, test responses, or data entry), observation, and

audiovisual recording. Data cannot be collected via physical procedures such as blood pressure monitoring, EEG, activity trackers (e.g. Apple Watch or FitBit), eye trackers, and blood draws.

XM4. Secondary Research for which consent is not required: Secondary research of identifiable private information or identifiable bio specimens.

[More Information](#)

XM5. Research and demonstration projects that are conducted or supported by a Federal department or agency, or otherwise subject to the approval of department or agency heads (or the approval of the heads of bureaus or other subordinate agencies that have been delegated authority to conduct the research and demonstration projects), and that are designed to study, evaluate, improve, or otherwise examine public benefit or service programs, including procedures for obtaining benefits or services under those programs, possible changes in or alternatives to those programs or procedures, or possible changes in methods or levels of payment for benefits or services under those programs.

[More Information](#)

XM6. Research involving taste and food quality evaluation and consumer acceptance studies. This category applies to Federal research only.

[More Information](#)

XM7. Storage or maintenance for secondary research for which broad consent is required: Storage or maintenance of identifiable private information or identifiable biospecimens for potential secondary research use if an IRB conducts a limited IRB review and makes the determinations required by 46.111(a)(8).

XM8. Secondary research for which broad consent is required. Research involving the use of identifiable private information or identifiable biospecimens for secondary research use.

More Information

*required

Do one or more of the above EXEMPT categories apply to your study?

✓ Yes

*required

Select ALL of the applicable EXEMPT categories, then click the right arrow to proceed.

1

2

✓ 3

4

5

6

7

8

No (continue to Expedited Categories)

Project Personnel

*required

Is this a student-led study?

Yes

✓ No

What kind of affiliation does the Principal Investigator (PI) have with Kean University?

*required

✓ Faculty

*required

✓ Tenured/Tenure-Track

Lecturer

Staff

Study Personnel

Note: If you cannot find a person in the people finder, please contact the IRB Office immediately.

*required

Principal Investigator

*The person listed as the PI will be required to certify submissions before they are sent to the IRB for review. They will also have edit access to this study, can create follow-on submissions after Initial approval, and will be included in study communications. **Note:** For student-led studies, list the faculty advisor as the Principal Investigator.*

Name: Iyadunni (Dunni) Adenuga
Organization: Department of Computer Science and Technology
Address:
Phone:
Email: iadenuga@kean.edu

*required

Primary Contact

Any people listed as a PC will have edit access to this study, can create follow-on submissions after Initial approval, and will be included in study communications.

Name: Kiana Becca Nunez
Organization: Users loaded with unmatched Organization affiliation.
Address:
Phone:
Email: beccanuk@kean.edu

Name: Iyadunni (Dunni) Adenuga
Organization: Department of Computer Science and Technology
Address:
Phone:
Email: iadenuga@kean.edu

Co-Principal Investigator(s)

*Any people listed as Co-Principal Investigators will have edit access to this study, can create follow-on submissions after Initial approval, and will be included in study communications. **They will also be required to certify the submission prior to IRB review.***

Other Personnel

Any people listed as Other Personnel will be able to view the study, but will NOT have edit access to the study nor be included in study communications automatically.

Name: Md Jonayed Hossain Chowdhury
Organization: Department of Computer Science and Technology
Address:
Phone:
Email: chowdmdj@kean.edu

Name: Kiana Becca Nunez
Organization: Users loaded with unmatched Organization affiliation.
Address:
Phone:
Email: beccanuk@kean.edu

Study Personnel Training Documentation

Upload documentation of any required training (e.g., CITI training) for each member of study personnel NOT affiliated with Kean University.

*required

Conflict of Interest

Do any of the study personnel have a financial conflict of interest related to this project?

Yes

✓ No

PROJECT BASICS

*required

Anticipated Start Date

06/22/2026

*required

Anticipated End Date

07/31/2026

*required

SUBJECT/PARTICIPANT POOL

*required

How many total subjects/participants will be enrolled in this study?

10

*required

What is the time commitment for each subject/participant involved in this study?

60 minutes

*required

SUBJECT/PARTICIPANT INFORMATION

Indicate the number of subjects/participants by age and, if applicable, by status for all categories that apply.

Newborn/Infants

Children (2-12 years old)

Adolescents (13-17 years old)

Pregnant Women

✓ Adults (18 or older)

*required

Number

10

Institutionalized (e.g., nursing home residents)

Incarcerated (prisoners)

Diagnosed with cognitive impairment, learning/language difficulty/mental illness

Other

*required

RESEARCH SITE(S)

Indicate where the project will take place. If available, attach any/all site approval letters.

NOTE: Some institutions require IRB approval prior to granting site permission. In these cases, site permission letters maybe be added letter as a follow-on submission.

✓ On-Site (Kean University)

*required

Indicate On-Site Locations

Remote via Kean's Microsoft Teams

✓ Off-Site

*required

Indicate Off-Site Locations

Remote via Kean Microsoft Teams

*required

Project Attributes

Check all that apply.

- ✓ Use of recruitment materials (e.g., flyers, emails, letters, advertisements, etc.)

[Recruitment Email.pdf](#)

- ✓ Questionnaires or surveys

*required

In-person

Phone

Mail

Email

- ✓ Online

[Study_Assist_Instructor_Evaluation_Surveys.pdf](#)

Other

- ✓ Interviews

In-person

Phone

- ✓ Video-conferencing

[INTERVIEW SCRIPT & THINK-ALOUD PROTOCOL.pdf](#)

Other

- ✓ Observation

Focus Groups

Administration of tests, inventories, self-reports, measuring instruments, etc.

- ✓ Photography, audio or video recording (NOTE: A separate Consent Form is needed for the participant to indicate consent for digital recording.)

Use of existing/secondary data

Medical procedures

- ✓ Consent Forms

[Informed Consent Form.pdf](#)

Assent Forms

- ✓ Consent Forms for audio/visual recording

[CONSENT TO AUDIO AND VIDEO RECORDING & TRANSCRIPTION.pdf](#)

✓ Other
*required

Please explain.

App Screenshots and Web Application Link

[chatHistory.png](#)

[ChatInterfaceCompleted.png](#)

[ChatInterfaceInitial.png](#)

[ChatInterfaceLearning.png](#)

[ChatInterfacePlan.png](#)

[favorites.png](#)

[onboarding.png](#)

[performance.png](#)

[profile.png](#)

[QuizFeedback.png](#)

[QuizOverlay.png](#)

[resetpassword.png](#)

[revision.png](#)

[signin.png](#)

[signup.png](#)

*required

Briefly describe your proposed research project, including your research goals/objectives.

Traditional AI chatbots often give answers too quickly, which can prevent people from learning through trial, effort, and problem-solving. This creates a situation where learners feel more confident, even though their actual understanding may not improve much. This research introduces a tool called Study Assist, an AI system designed to help people learn more effectively. Instead of simply giving answers, it guides users through important learning steps like planning, checking their progress, and reflecting on what they learned. The system includes features that encourage active learning, such as editable study plans, progress checkpoints with reflection questions, and even disabling copy-paste so users must think and recall information themselves. In addition to the student version, the system also includes an instructor version, which helps instructors monitor student progress, review learning behaviors, and better integrate the tool into classroom instruction. This instructor version supports classroom management and allows educators to guide students more effectively throughout the learning process.

The goal of this research is to evaluate how computer science instructors interact with, interpret, and leverage AI-generated student learning analytics to guide their instructional workflows.

The following research questions guide our work:

- RQ1: How does an Expert-in-the-Loop architecture reshape the design requirements of an LLM-based educational system?
- RQ2: What dimensions of expert oversight do instructors value when integrating AI tutoring platforms into their courses?
- RQ3: How do instructors envision using AI-generated learning data to inform their teaching practices?

*required

Explain how you will recruit subjects into the proposed research project. Describe when and where recruitment will be conducted and methods of recruitment to be used (e.g. flyers, email, social media, face-to-face).

Participants will be recruited from the faculty pool within the Department of Computer Science and Technology at Kean University. We are targeting a cohort of 5 to 10 professors or instructional staff who possess experience teaching introductory programming courses (such as Java-based CPS 1231).

Recruitment will be conducted between June 22nd and July 30th, 2026, using purposeful institutional outreach. The student researcher will contact prospective faculty members directly via official university email using an IRB-approved outreach letter on Kean University letterhead. The email will contain a

secure scheduling link for faculty members who choose to volunteer for an evaluation session.

*required

Briefly describe the nature of the involvement of the human subjects (e.g., observation of postsecondary student behavior in the classroom, personal interviews, mailed questionnaire, survey, chart review, etc.).

Participants will take part in a single, individual 60-minute remote user study session conducted synchronously over Microsoft Teams. After providing digital informed consent on an approved form bearing Kean letterhead, participants will engage in an interactive user evaluation following a Within-Subject Design and a Think-Aloud Protocol.

Instructors will log into a web-based prototype of the Study Assist instructor dashboard populated with a standardized simulation of 20 distinct student personas representing varying capabilities, programming backgrounds, and behavioral attributes over a 12-week semester layout. While verbalizing their thought processes, instructors will complete structured tasks, including:

1. Evaluating high-level class performance summaries.
2. Navigating a mid-semester snapshot (Week 4) to identify a student flagged as "About to Fail" due to hitting automated milestone retry limits and reflection blockages.
3. Navigating an end-of-semester snapshot (Week 10) to evaluate longitudinal concept mastery and student reward points.
4. Auditing assessment tracking metrics to scan for potential LLM errors or hallucinations.

Following the hands-on interaction, participants will fill out an online Qualtrics survey incorporating an adapted System Usability Scale (SUS) and answer qualitative, semi-structured interview questions focusing on dashboard utility, classroom integration prerequisites, and pedagogical control mechanisms. The Microsoft Teams sessions will be audio and video recorded solely for content analysis and transcription accuracy, subject to secondary participant consent.

*required

Describe the setting (e.g., a classroom) and the location (e.g., name of school) where the research will be conducted.

NOTE: If research is to be conducted at another institution or facility, a signed copy of the site permission letter from that institution authorizing the researcher to collect data on its grounds must be submitted to the IRB prior to initiation of your project, even if IRB approval is granted.

The research will be conducted completely online and remotely via secure, institutional Kean University Microsoft Teams video conferencing links and cloud-hosted web prototype applications. Because the software prototype is web-accessible and the participant pool consists entirely of adult professional educators engaging via remote computers, no physical classroom or lab space is utilized. Post-task questionnaires and feedback forms will be securely served online using the Qualtrics platform.

*required

Explain how records will be kept, who will have access to the data, and how the data will be stored.

Quantitative interaction metrics and qualitative questionnaire responses will be collected through the secure Qualtrics platform. These files, alongside remote session audio/video files, text notes, and interview transcripts, will be uploaded directly to the Faculty Principal Investigator's secure, password-protected Kean University OneDrive cloud storage account. No identifiable research materials will be stored on local unencrypted drives.

Access to the raw data is strictly confidential and restricted exclusively to approved project personnel (the Faculty PI and the student Co-Investigator) to assist with qualitative content analysis and thematic coding. The Kean University Institutional Review Board (IRB) administrative staff will be granted access upon request if required for compliance verification.

*required

Explain how the data or records will be deleted or destroyed.

In accordance with academic data preservation standards, all digital records, survey exports, transcripts, and multimedia interaction recordings will be preserved securely on the PI's institutional Kean OneDrive account for a mandatory retention period of 5 years following the formal conclusion of the study. Once the 5-year mark is reached, the student investigator or faculty PI will permanently delete all folders, datasets, and multimedia files from the cloud repository and empty the digital trash caches to ensure complete destruction.

*required

Explain why you think this protocol should be considered Exempt. Address all known or potential risks to subjects/participants.

This protocol qualifies for Exempt Status under Category XM3 (Benign Behavioral Interventions). The participants are fully competent adult professionals who are prospectively volunteering to complete a brief, online educational software walkthrough and subsequent usability interview. The study is harmless, painless, and entirely non-invasive.

Supporting Documents

Attach any additional documents to be used as part of this study.